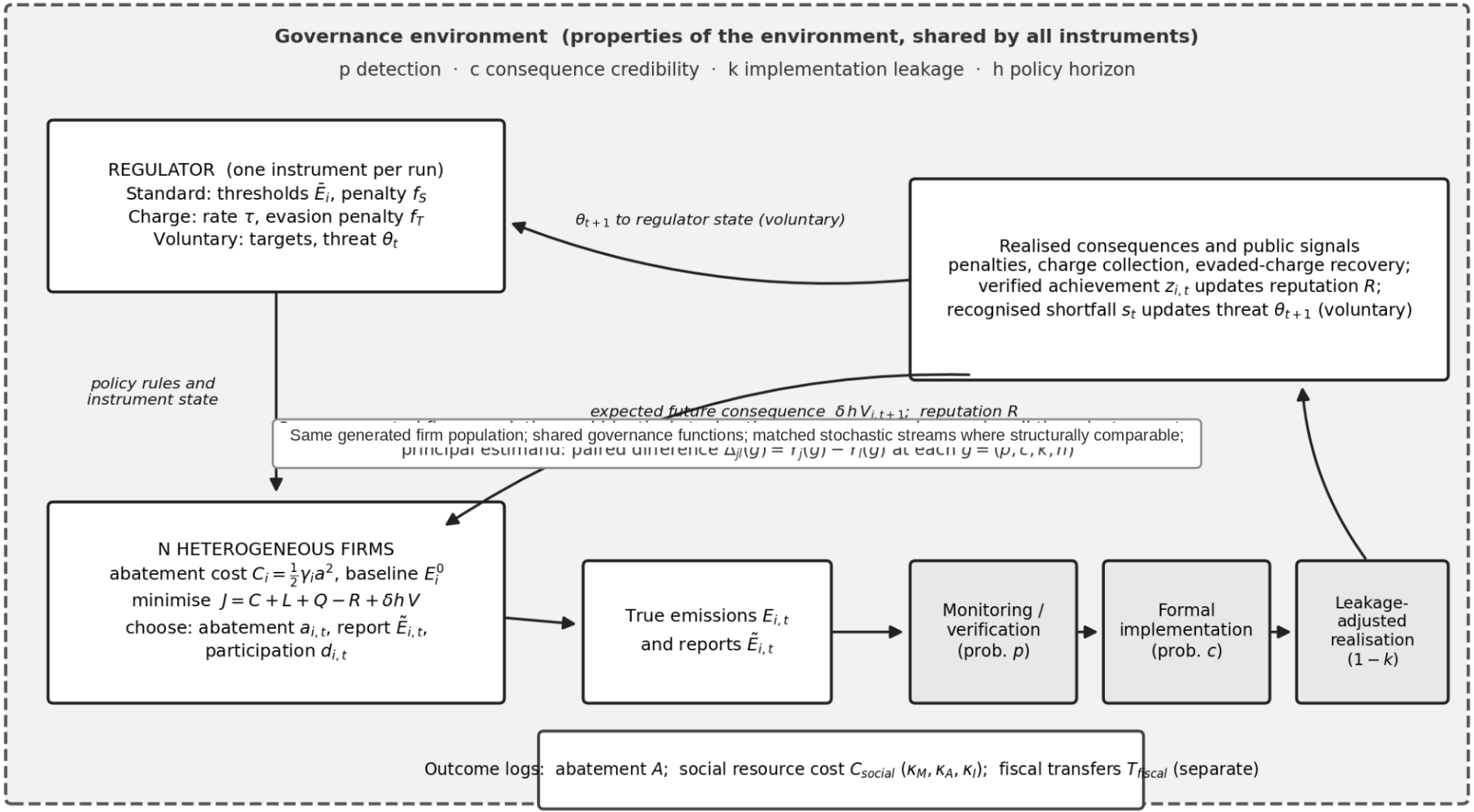

Governance environment (properties of the environment, shared by all instruments)
p detection · c consequence credibility · k implementation leakage · h policy horizon
REGULATOR (one instrument per run)
Standard: thresholds $\bar{E}_i$, penalty f_S
Charge: rate τ, evasion penalty f_T
Voluntary: targets, threat θ_t
θ_{t+1} to regulator state (voluntary)
Realised consequences and public signals
penalties, charge collection, evaded-charge recovery;
verified achievement $z_{i,t}$ updates reputation R;
recognised shortfall s_t updates threat θ_{t+1} (voluntary)
policy rules and instrument state
expected future consequence $\delta h V_{i,t+1}$; reputation R
Same generated firm population; shared governance functions; matched stochastic streams where structurally comparable;
principal estimand: paired difference $\Delta_{jl}(g) = Y_j(g) - Y_l(g)$ at each $g = (p, c, \kappa, h)$
N HETEROGENEOUS FIRMS
abatement cost $C_i = \frac{1}{2}\gamma_i a^2$, baseline E_i^0
minimise $J = C + L + Q - R + \delta h V$
choose: abatement $a_{i,t}$, report $\tilde{E}_{i,t}$,
participation $d_{i,t}$
True emissions $E_{i,t}$
and reports $\tilde{E}_{i,t}$
Monitoring / verification (prob. p)
Formal implementation (prob. c)
Leakage-adjusted realisation ($1-k$)
Outcome logs: abatement A; social resource cost C_{social} ($\kappa_M, \kappa_A, \kappa_I$); fiscal transfers T_{fiscal} (separate)